

FairLens — Bias Mitigation Report

 16 Mar 2026

 Target: **income**

 Protected: **race**

 Privileged: **White**

 Rows: **48,842**

EXECUTIVE SUMMARY

-0.1014 Baseline SPD	-0.0301 Final SPD (TabNet)	70.3% Bias Reduction	0.8471 Best Accuracy
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FULL PIPELINE RESULTS

Model	Stage	Accuracy	SPD	SPD Status	DI	DI Status	AOD	EOD
RF	Baseline	0.8589	-0.1010	✗ Unfair	0.5296	✗ Unfair	-0.0483	-0.0581
	Reweighting	0.8582	-0.0956	✗ Unfair	0.5581	✗ Unfair	-0.0395	-0.0447
	Reweigh+ADB	0.8497	-0.0207	✓ Fair	0.8817	✓ Fair	0.0654	0.1140
	★ Full Pipeline	0.8493	-0.0320	✓ Fair	0.8174	✓ Fair	0.0391	0.0662
XGBoost	Baseline	0.8757	-0.1021	✗ Unfair	0.5309	✗ Unfair	-0.0456	-0.0577
	Reweighting	0.8733	-0.0604	✗ Unfair	0.7109	✗ Unfair	0.0148	0.0312
	Reweigh+ADB	0.8497	-0.0303	✓ Fair	0.8455	✓ Fair	0.0505	0.0870
	★ Full Pipeline	0.8497	-0.0308	✓ Fair	0.8431	✓ Fair	0.0502	0.0870
TabNet	Baseline	0.8517	-0.0887	✗ Unfair	0.5596	✗ Unfair	-0.0430	-0.0574
	Reweighting	0.8484	-0.0352	✓ Fair	0.8108	✓ Fair	0.0392	0.0694
	Reweigh+ADB	0.8473	-0.0076	✓ Fair	0.9596	✓ Fair	0.0828	0.1351
	★ Full Pipeline	0.8471	-0.0301	✓ Fair	0.8397	✓ Fair	0.0366	0.0554
LightGBM	Baseline	0.8776	-0.1001	✗ Unfair	0.5379	✗ Unfair	-0.0437	-0.0567
	Reweighting	0.8740	-0.0620	✗ Unfair	0.7073	✗ Unfair	0.0123	0.0265
	Reweigh+ADB	0.8518	-0.0617	✗ Unfair	0.6814	✗ Unfair	-0.0024	0.0035
	★ Full Pipeline	0.8400	-0.0319	✓ Fair	0.8056	✓ Fair	0.0458	0.0893

SMOTE OVERSAMPLING RESULTS

Variant	Accuracy	SPD	DI	AOD	EOD
Baseline (no SMOTE)	—	-0.1014	0.6006	—	—
Borderline	0.8509	-0.1143	0.5165	-0.0669	-0.0853
KMeans	0.8526	-0.1099	0.5300	-0.0569	-0.0675
ADASYN	0.8498	-0.1158	0.5104	-0.0669	-0.0825
Standard	0.8526	-0.1099	0.5300	-0.0569	-0.0675

FEATURE IMPORTANCE (TOP 10)

Feature	Importance	Visual
fnlwgt	17.55%	
age	15.60%	
capital_gain	12.06%	
relationship	9.91%	
education_num	8.63%	
hours_per_week	8.42%	
marital_status	6.83%	
occupation	6.57%	
workclass	3.78%	
capital_loss	3.77%	

CONFIGURATION

Parameter	Value
Models	RF, XGBoost, TabNet, LightGBM
Mitigation Steps	Reweighting → ADB → CEO
SMOTE Variants	Borderline, KMeans, ADASYN, Standard
Test Split	30%
Feature Count	13
Total Rows	48,842
Class Distribution	0: 37,155 · 1: 11,687

Generated by FairLens · Fairness detection and mitigation framework · Fair zones: |SPD| < 0.05 · DI ∈ [0.8, 1.25] · |AOD| < 0.05 · |EOD| < 0.05